LIFELINE - Approach Note (Group 76, seed s = 76)

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1. From story to numbers

Case study	Choices (measured over our cases)	Story sentence that justifies them
1 Delta Flood	Planar-like geometric network (spanning tree to the nearest earlier point, extra roads only between near neighbours), E in [1.1V, 1.6V]: 2E/V = 2.31-2.96. 45-60% of locations get penalty 2/4/6 (measured 43-55%). Minutes x1.3. 3-5 waves that close roads touching high-penalty locations first. Food/water forced to 58-72% (72-100%). Food and water are bulky (4-20 kg) and cheap.	"A flat delta of small villages joined by raised roads. Roads go under water in waves. Food and water matter most."
2 Mountain Earthquake	E in [V, 1.3V]: 2E/V = 2.08-2.50, so the network is almost a tree. Minutes x1.4. Medicine/tools forced to 48-62% (73-92%). Alarms: 45% fall in bursts around aftershock centres, and burst alarms get priority >= 6. Mean priority 7.8-8.2.	"A few roads and almost no alternatives. The injured need medicine and tools. Bursts of urgent calls follow each aftershock."
3 Coastal Cyclone	Shelters ceil(V/4) + up to V/10 (27-44%). 3-5 waves. Windows last only 5-75 min and start around 2..n/6 surge peaks (sigma 35 min), so a window overlaps 3.3-18.4 others on average on S/M and 171 on the L case (we require >= 2.5). Radio: 30% of message symbols replaced by uniform noise over all 38 symbols, so 37-38 symbols are used.	"Storm surge closes roads in waves. Rescue windows are short and overlap. The radio is full of noise."
4 Urban Flash Flood	E in [1.85V, 2.6V]: 2E/V = 3.75-5.11, max degree up to 14. Minutes x0.7 (short city blocks). Alarm rate 1.55-1.75 per tick on M/L (measured 1.56-1.73). Rare crates forced to 12-20% (15-53%).	"A dense city with many alternative routes. Calls arrive faster than anyone can answer them. Rare equipment is available."
5 Landslide Corridor	A highway path from the HQ through 45-60% of the locations, plus spurs (villages). Extra loops only join a spur to a location at most 2 highway positions away, so they never shorten the highway, and every degree is capped at 4. 2E/V = 1.78-2.40; the HQ's farthest location is P-1 >= floor(V/4) roads away (e.g. 356 on V = 769). Wave 1 always contains a highway bridge whose loss cuts off a shelter or hospital, which becomes target_0.	"A single mountain highway along a gorge. A collapse cuts off whole communities."
6 Outbreak Camp	Water/medicine forced to 58-72% (71-90%). 55-70% of crates get expiry 6..300 (measured 65-79%). The beacon phrase is weighted x6, and every message is followed by an idle run of 3-9 '-', so '-' is 31-44% of the log.	"Medicine and water count and supplies spoil. Radio beacons repeat the same messages."

Common choices. Crate profiles: food 4-20 kg/value 10-55, water 6-20/15-65, medicine 1-8/40-100, tools 3-18/25-85, rare 1-12/60-100. Capacity is 20-50% of total weight, so it always binds. Road minutes are proportional to geometric length plus noise, and cable is 0.8-2.2x the minutes (terrain). Pipeline dimensions mix narrow 2-10 (25%), 10-60 and 60-100 stages so that the bracketing matters. Drum prices are about unit x j with premium "standard" lengths, saturating below 500. Sites are 80% Gaussian village clusters, so close pairs exist. Ration values are 5-300 (S) or 100-1000. Sensor cover size k is drawn from [n/4, n/2].

2. Planted objects and claims (correct by construction)

- **Tour.** Nearest-neighbour tour from site 0 (ties go to the smaller site_no), then deterministic best-improvement 2-opt with site 0 fixed. 2-opt never lengthens a tour, so the planted tour is never longer than the NN tour. On L/X it is 6-17% shorter, so it also beats NN under any tie rule. tour_length is measured from the stored tour.
- **Sensor cover.** The cover C (k nodes) is drawn first. Every edge joins a non-cover node to C, or joins two nodes of C, so C touches every edge by construction. We add edges until there are at least n-1.
- **Safety assignment.** The 0/1 assignment is drawn first. Then we draw random rules over three distinct switches and keep a rule only if the assignment satisfies it (duplicates are rejected).
- **Rations.** The planted subset (2..q/2 items) is drawn first, and rations_target is set to its sum.
- **Radio.** Pattern counts are computed on the joined log with overlaps. An "occurring" pattern is copied from the log. A "missing" pattern is a one-symbol mutation of a log substring, checked to occur 0 times. edit_bound is the exact Levenshtein distance (DP), so it lies between the true distance and len(a)+len(b).
- **Claims (tier S).** The generator uses exhaustive search: knapsack over all 2^n subsets (meet-in-the-middle enumeration), all compatible window subsets by DFS, all integer partitions of L for the drum, all (n-1)!/2 tours, and covers tried by increasing size. The validator recomputes every claim with a different exact method: O(nW) DP, earliest-finish greedy plus weighted-interval DP, O(L^2) rod DP, Held-Karp, and branch-and-bound vertex cover. The two agree on all 20 tier-S cases.
- **Variety.** Variety 1: tier-S cases with k mod 3 != 2 are redrawn until ratio-greedy is not optimal (14 forced, 15 hold). Every tier-S crate list has distinct value/kg, so the greedy order is unambiguous. Variety 2: tier-S cases of studies 1, 3, 4 and 6 are redrawn until some S/H location's fewest-roads routes all cost more than its cheapest route, and that location becomes target_0 (13 forced, 15 hold). Variety 3: a seeded shuffle gives the 26 M/L cases 7 / 12 / 7 rule ratios in [2.6, 3.3] / [3.7, 4.6] / [5.0, 5.9].

Variety 4: serials come from a per-case block of 1.8M numbers, so they are distinct across the workbook. Each draw that breaks a rule is rejected and redrawn with attempt+1, so the result stays deterministic.

3. The traps

Trap	What we built	Why it is hostile	Check
46 Dock order (story 5)	1,920 crates in dock order: weight non-decreasing, value non-increasing (hence value/kg non-increasing), expiry non-increasing, serial increasing. All 5 fields are monotone, with heavy duplicate keys (weights 1..20).	Sorted and reverse-sorted input sends first/last-pivot quicksort to O(n^2) and defeats algorithms that assume random order. Many equal keys break naive partition. Also the largest knapsack (n x W = 9.8M cells).	V8: counts monotone fields (value/kg by exact cross-multiplication), needs >= 3. V2: 1,500-3,000 crates.
47 Best-looking crate (story 6)	16 crates: 15 crates of 20 kg worth 80-100 (at most 5/kg) and one 1 kg decoy worth 10 (10/kg). Capacity 40.	Greedy takes the decoy, then one heavy crate; the 19 kg left fits nothing, giving 110. The optimum is the two best heavy crates, 199. Greedy gets 55.3%.	V8: exact greedy (ties would be reported) vs DP optimum, 10 x greedy <= 6 x opt. Also checks 12-60 crates.
48 Shortest looks cheapest (story 1)	1,630 locations. A cheap backbone road (1-4 min, penalty 0/1) of about 1,000 locations, with spurs: the HQ's eccentricity is 803 roads. A 35-60-road causeway (50-60 min, penalty 4/6) joins the HQ to a hospital about 300 roads down the backbone.	The fewest-roads route to target 540 costs 3,659 vs 753 for the cheapest (4.9x). BFS or hop-count heuristics pick the dear route. Paths of 800+ roads strain recursive search (Python's default limit is 1,000 frames).	V8: double-sweep BFS >= 500 (exact all-pairs fallback). BFS-layer DP vs Dijkstra: 2 x fewest-roads cost >= 3 x cheapest for some target.
49 Near miss (story 2)	54,004 symbols. 95% of it is one symbol Q, in three long runs, separated by two noise bursts with no Q. Every pattern is Q^(m-1) followed by another symbol.	The first m-1 symbols match at about 95% of all start positions for all six patterns, yet only two patterns occur, once each. Naive search does Theta(nm), about 2 x 10^6 comparisons for m = 40. Huffman gets one symbol with 95% frequency.	V8: best block of length <= 8 (most frequent p-gram, non-overlapping cover) >= 90%. Near-miss prefixes counted with KMP; needs >= 4.

The validator recomputes all of this from raw rows and never imports generator code. `validator/tests/` holds 4 valid copies and 27 broken copies (12 breaks of case 76-06, 10 of 76-29, 3 trap breaks, and 2 whole-workbook breaks). validation_report.txt shows all 50 cases PASS and every broken copy rejected by the intended rule.

4. What will be hard

Case study	Hardest decisions and the data property behind each
1 Delta Flood	**B route.** Step cost minutes + penalty(v) is asymmetric, 43-55% of locations carry penalty >= 2, and 3-5 waves remove exactly the low-lying roads, so the fewest-roads route is often dearer (76-00, 76-12, 76-18). **A load.** Heavy, cheap food/water with near-equal value/kg; greedy is suboptimal (e.g. 473 vs 483 in 76-00).
2 Earthquake	**D queue.** Mean priority about 8 with urgent bursts, so many alarms tie at priority 9-10; tie order (tick, then alarm_id) must be stable. **F cable.** 2E/V of only 2.1-2.5 and closures split the network: the MST must cover only the HQ's component (a spanning forest otherwise).
3 Cyclone	**C windows.** 5-75-minute windows overlap 3-171 others on average and many touch (finish = start), so < vs <= changes the answer; value maximisation needs DP, not greedy. **E radio.** 37-38 near-uniform symbols leave Huffman little to gain (4.6-4.9 bits per symbol vs 6 for a fixed code), and pattern search runs through noise.
4 Urban	**D queue.** 1.56-1.73 alarms arrive per tick but one is released, so the backlog peaks at 0.36-0.47n (861 alarms in 76-45); a linear scan becomes O(n^2). **B route.** Dense graphs (2E/V up to 5.1, degree up to 14) with many equal-cost alternatives; the heap holds up to 2E entries.
5 Landslide	**B reach.** Waves cut target_0 off, so "no route" must be reported; corridors up to 356 roads deep (291 in trap 46) strain recursive search. **F cable.** The MST must span only the locations still reachable after the collapse, not all V.
6 Outbreak	**A load.** 65-79% of crates spoil by hour 300, so "the values of the day" change, and value/kg misleads (trap 47 at 55%). **E radio.** The log is skewed (filler 31-44%), so Huffman gives a 1-2 bit code to the filler, and beacons produce many overlapping occurrences.

For module H across all studies: rule ratios near 4.26 (the 3.7-4.6 bucket) are the hardest region for DPLL, and L-size tours (200-320 sites) and covers (k up to 65) are far beyond exact search. There we will report heuristic answers with bounds against the planted objects.